AI FIRST NATIONS / MANGROVE AI

Binary Mangrove Detection HD Model Evaluation Specification

Independent Jupyter Notebook assessment | 100-mark rubric | Application-readiness verification

Specification	Requirement
Model	EfficientNet-B0 (2-class binary classifier)
Primary output	Reproducible .ipynb notebook with quantitative evidence
Decision	Evidence-based deployment recommendation: Ready / Experimental / Not Ready
Assessment status	Proposed HD-standard rubric; not an official RMIT marking scheme

1. Purpose and evaluation questions

Develop a research-quality evaluation of the Binary Mangrove Detection model already used in the Mangrove AI pipeline. The aim is to measure genuine predictive performance on independent drone imagery and establish whether the saved weights and preprocessing operate correctly in Flask. High reported accuracy alone does not establish field readiness.

- Can the model distinguish mangrove from non-mangrove imagery on genuinely unseen source images and survey sites?
- How does it behave on realistic negatives (other coastal vegetation, water, mud, shadows and non-mangrove trees)?
- Do tiled-image predictions and aggregation produce reliable full-image decisions?
- Can the exact model and inference configuration be reproduced in the deployed Flask application?

2. Required developer deliverables

Deliverable	Required evidence
Notebook	Binary_Mangrove_Evaluation.ipynb; executed outputs and narrative interpretation
Data audit	Manifest with original image IDs, site/survey identifiers, labels, tile lineage and splits
Metrics	Machine-readable CSV/JSON metrics, confusion matrix, per-image predictions
Visual evidence	Learning curves, calibration, PR/ROC where valid, failure cases, tiled-image examples
Application tests	Actual saved model loaded and compared with Flask inference
Handover	Requirements/config, seed, model checksum, reproducible run instructions

3. Mandatory notebook sections

01. Research design

Define scope, labels, intended use, hypotheses, evaluation unit (tile vs source image) and acceptance criteria.

02. Dataset audit

Inventory original images, tiles, site/survey IDs, annotation provenance, label balance, resolution, quality and duplicates.

03. Split integrity

Group by source image; where feasible hold out entire survey/site. Demonstrate no source/tile overlap across partitions.

04. Model reproducibility

Record architecture, classifier head, weight hash, class-index mapping, transforms, batch size, device, seeds and package versions.

05. Core predictive metrics

Calculate all mandatory classification metrics on a locked independent test set; explain denominators and aggregation.

06. Tile-to-image analysis

Evaluate per-tile and full-source-image predictions separately; test aggregation and edge/coverage behaviour.

07. Robustness and subgroup analysis

Report performance by site, canopy, acquisition conditions and difficult cases when labels/sample sizes permit.

08. Controlled improvements

Compare original baseline against justified changes using validation data only; do not tune on test data.

09. Flask end-to-end verification

Test actual model loading, tiler, output schema, runtime, memory, failure states and prediction consistency.

10. Conclusions and readiness

Summarise evidence, uncertainty, error analysis, known limits, acceptance checks and final recommendation.

4. HD marking rubric (100 marks)

Criterion	Marks	HD standard
Research questions and evaluation design	5	Specific, measurable, deployment-linked questions and valid methodology
Dataset audit and annotation quality	10	Comprehensive provenance, distributions, duplicate/label review
Leakage prevention and splits	15	Documented group-wise splits; site holdout where possible; verified isolation
Reproducibility and model setup	10	Exact saved weights, transforms, class order, seeds and environment
Classification metrics and calibration	20	Correct class-wise metrics, confusion matrix, confidence calibration and interpretation
Robustness, confounding and failure analysis	15	Realistic negatives/shift tests, error taxonomy and subgroup limitations
Controlled improvement experiments	10	Fair baseline comparison; validation-only tuning and reasoned conclusions
Flask integration and operational tests	10	Verified production-path predictions, latency, resource and error tests
Notebook clarity and conclusions	5	Well-presented, executable, critical and evidence-based

Grading descriptors: HD 80-100 = rigorous and reproducible with independent testing and critical analysis; D 70-79 = comprehensive with minor gaps; C 60-69 = adequate metrics but limited robustness; P 50-59 = basic evaluation; below 50 = unreliable or materially incomplete.

5. Mandatory quantitative metrics

Metric	Mandatory interpretation
Confusion matrix (TN, FP, FN, TP)	Use Mangrove as the positive class; define labels explicitly
Precision / Recall / F1 (each class)	Report class support; foreground false negatives and false positives
Balanced accuracy / macro F1	Avoid hiding class imbalance behind overall accuracy
ROC-AUC and PR-AUC	Use continuous Mangrove scores; report class prevalence for PR-AUC
Sensitivity / specificity / false-positive rate	Quantify operational screening trade-offs
Brier score / ECE / reliability plot	Test confidence calibration; high softmax score is not proof of correctness
Full-image detection accuracy	Report independently of tile-level metrics
Latency and peak memory	Measure on the actual target hardware and file sizes

Formulas: Precision = TP/(TP+FP); Recall = TP/(TP+FN); F1 = 2PR/(P+R); Accuracy = correct predictions / total predictions. For three classes, compute metrics per class and report macro averages. Undefined metrics must

be reported as undefined, not silently replaced with zero.

6. Evaluation experiments - mandatory

A. Reproduce the inherited baseline

Load bestBinary.pth; reproduce current preprocessing and source-level inference without retraining. Compare reported prior-team accuracy with new independent measurements.

B. Realistic negative challenge set

Evaluate water, mudflats, non-mangrove vegetation, other trees, urban/coastal structures and ambiguous canopy. Report FP rate and representative errors.

C. Site and source generalisation

Hold out source images and, where feasible, an entire site/survey. Compare in-domain vs held-out results; discuss uncertainty.

D. Tile aggregation and threshold

Compare tile-level predictions with source-image aggregation (current probability averaging); use validation set to select thresholds. Examine mixed scenes with small mangrove regions.

E. Calibration and operating point

Plot reliability and PR/ROC curves; choose a screening threshold based on acceptable missed-mangrove vs false-alarm trade-offs.

F. Controlled improvement

Compare at least two justified improvements, e.g. representative negatives, calibration, preprocessing or aggregation, with a locked final test set.

7. Data protocol and leakage controls

- Create a manifest: source_image_id, tile_id, site_id, survey_id, acquisition metadata if permitted, label, split and annotation provenance.
- Split at source-image level before tiling; preferably test cross-site or cross-survey generalisation. Do not split sibling tiles across train and test.
- Review exact and near-duplicate imagery, overlapping orthomosaic areas, and temporally adjacent frames that may leak scene identity.
- Freeze a held-out test set and preserve an experiment log. Threshold tuning, calibration and model selection must use validation data only.
- Report test sample sizes and class support at original-image level, not merely the number of highly correlated tiles.
- For confidence intervals, bootstrap by independent source image or site/survey (not tile) where sample size permits.
- Respect Indigenous data governance: keep survey data within approved storage and do not upload to unapproved annotation/cloud services.

8. Performance targets and acceptance gates

The following are proposed engineering targets, not promises or universal scientific thresholds. Confirm them with the project team before running the locked final test. A model may receive an HD evaluation even when it fails these targets, provided the methodology and conclusions are rigorous.

Measure	Proposed target
Mangrove recall (source-image level)	>= 95%
Mangrove precision (source-image level)	>= 90%
Macro F1 (source-image level)	>= 0.90
Held-out site evaluation	Completed; limitations explicitly stated
Calibration	Reliability plot and score reported
Flask critical integration checks	All pass

A deployment recommendation must also consider independent sample size, false-result consequences, site coverage and runtime. The model is not production-validated solely by reaching the numerical targets.

9. Required plots and examples

- Source-level train/validation/test distributions and class counts
- Representative original images and their tile lineage
- Confusion matrix (raw counts and normalised)
- Class-wise precision/recall/F1 with support
- Confidence calibration / reliability diagram
- ROC/PR curves where statistically valid
- Correct high-confidence and incorrect high-confidence examples
- Per-site or per-condition results with sample sizes
- Baseline vs improved-model comparison
- Full-image versus tile-level prediction comparison
- Flask output screenshot paired with notebook output and numerical checks

Every important figure must have a short interpretation: what it shows, why it matters, and what uncertainty or limitation remains.

10. Flask application verification

Test the actual application inference path, not just a standalone model call: Upload -> image validation -> shared image_tiler.py -> preprocessing -> saved EfficientNet-B0 weights -> tile aggregation -> response object -> results page. The species classifier should run only when the upstream binary result permits it under the application policy.

- Exact model checkpoint and class ordering are identical between notebook and Flask.
- Input transforms (resize to 224x224, normalisation, channel handling) and tile settings are consistent.
- Predicted probabilities and image-level labels match within a documented floating-point tolerance.
- Large, small, edge-sized, mixed-content and corrupted images are handled predictably.
- Run cold-start and warm-inference latency plus peak RAM; state hardware, batch size and image dimensions.
- When a model is unavailable, the app shows a safe unavailable state rather than invented output.
- For species: verify upstream non-mangrove rejection and unsupported/mixed-category limitations.
- Provide a pass/fail table with test ID, input, expected, observed, evidence and issue status.

11. Final results summary - required template

Metric	Baseline	Final	Target / criterion	Pass?
Mangrove precision	TBD	TBD	Pre-agreed	TBD
Mangrove recall	TBD	TBD	Pre-agreed	TBD
Non-mangrove specificity	TBD	TBD	Pre-agreed	TBD
Macro F1	TBD	TBD	Pre-agreed	TBD
Balanced accuracy	TBD	TBD	Pre-agreed	TBD
PR-AUC	TBD	TBD	Pre-agreed	TBD
Brier / ECE	TBD	TBD	Pre-agreed	TBD
Full-image accuracy	TBD	TBD	Pre-agreed	TBD
Held-out site performance	TBD	TBD	Pre-agreed	TBD
Flask integration	TBD	TBD	Pre-agreed	TBD

12. Non-negotiable HD integrity requirements

- No leakage between train and test at source-image level.
- No accuracy claim without independent labelled ground truth and denominators.
- No threshold/model tuning using final test data.
- No reporting only overall accuracy while hiding minority-class errors.
- No unqualified claims about species identification, health, carbon or field reliability.
- No claims of application readiness without running the actual Flask integration checks.
- No invented metrics, fabricated screenshots, cherry-picked results or omitted failure cases.
- Notebook must execute top-to-bottom with clear paths, dependency versions and reproducible outputs.

13. Final developer sign-off checklist

[ ] Independent source-level test set verified

[ ] Annotation/label mapping audited

[ ] All mandatory metrics and calibration completed

[ ] Failure cases and realistic field conditions analysed

[ ] Controlled experiments compared fairly

[ ] Full-image and tile-level predictions differentiated

[ ] Flask inference and runtime tested

[ ] Outputs saved as figures + CSV/JSON

[ ] Notebook reproducible in clean environment

[ ] Deployment recommendation justified

14. Final assessment statement

Conclude with: (1) measured independent test performance; (2) performance variation across conditions/sites; (3) main failure modes and risks; (4) agreement between notebook and deployed Flask inference; (5) a justified status of Ready for defined use, Experimental/Conditional, or Not Ready. Distinguish model confidence from verified real-world accuracy.